

		v_r
11	A	$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT \rightarrow c_{rms} = \sqrt{\frac{3kT}{m}}$ $\frac{c'_{rms}}{c_{rms}} = \sqrt{\frac{T'}{T}} \rightarrow \frac{0.8c_{rms}}{c_{rms}} = \sqrt{\frac{T'}{295}}$ $T' = 188.8 \text{ K} = \mathbf{-84.4^\circ \text{C}}$
12	C	$\mathcal{E} = (1.5)(0.5) \text{ A} = \mathbf{0.75 \text{ V}}$